

Homework 4

- 1) X is r.v. whose possible values are $0, 1, 2, \dots$
and has CDF F

$$G(x) = P(X < x)$$

$$G(x) = F(x) - P(X = x) = F(\lceil x \rceil)$$

- 2) n eggs, each hatches w/ prob p (independently)
each chick survives w/ prob r

$H \rightarrow$ r.v. of hatched eggs

$S \rightarrow$ r.v. of surviving chicks

$$P(H) \sim \text{Bin}(n, p)$$

$$P(S) \sim \text{Bin}(n, pr)$$

- 3) Indicator B after first child is girl
" " " " " " boy

$\hookrightarrow$ each are Geometrically distributed

$$E(\text{Geom}(\frac{1}{2})) + 1 + 1 = 3$$

$\uparrow$ $\uparrow$ $\uparrow$
 # of 'failures' new gender baby first child

- 4) k distinguishable balls placed in
 n distinguishable boxes
all possibilities equally likely

Let indicator $E_i \begin{cases} \rightarrow 1 & \text{box } i \text{ empty} \\ \rightarrow 0 & \text{box } i \text{ not empty} \end{cases}$

$$n E(E_1) = n \cdot 1 \cdot \left(\frac{n-1}{n}\right)^k$$

5) a) m women n men
 $X \sim \text{Bin}(m, p_1)$ $Y \sim \text{Bin}(n, p_2)$

interested in null hypothesis $p_1 = p_2$

$X + Y = r$ → condition on this

	Disease	No Disease	
W	x	$m - x$	m
M	$r - x$	$n + x - r$	n
	r	$m + n - r$	$m + n$

$$\begin{aligned}
 P(X = x \mid X + Y = r) &= \frac{P(X + Y = r \mid X = x) P(X = x)}{P(X + Y = r)} \\
 &= \frac{P(Y = r - x) P(X = x)}{P(X + Y = r)} \\
 &= \frac{\binom{n}{r-x} p^{r-x} (1-p)^{n-r+x} \binom{m}{x} p^x (1-p)^{m-x}}{\binom{n+m}{r} p^r (1-p)^{m+n-r}} \\
 &= \frac{\binom{r-x}{x} \binom{m}{x}}{\binom{n+m}{r}}
 \end{aligned}$$

$$= H_{\text{Green}}(n, m, r)$$

b) If $p_1 = p_2$ and we know $X + Y = r$, it makes sense that the probability disappears when calculating the CDF bc we know how many folks are diseased and not diseased. we can just count the # of combinations now

6) a) For an list size n , there are $n-1$ sweeps
 and $1 + 2 + \dots + n-1 = \frac{n(n-1)}{2} = \binom{n}{2}$ possible
 inversions

Indicator $S_i \begin{cases} \rightarrow 1 & i^{\text{th}} \text{ pair inverted} \\ \rightarrow 0 & \text{" " NOT inverted} \end{cases}$

$$\mathbb{E}(S_i) = \binom{n}{2} \cdot 1 \cdot \frac{1}{2} = \binom{n}{2}$$

b) upper bound: there are $\binom{n}{2}$ possible pairs
 lower bound: there are $\frac{1}{2}\binom{n}{2}$ expected inversions
 and not all comparisons result
 in an inversion

7) a) $X_1, X_2, \dots$ iid

The prob of " i th jumper sets record" is
 independent of " j th jumper sets record"

the former prob is $\frac{1}{n}$
 latter prob is $\frac{1}{n}$

Intuitively, the jump height of previous jumpers
 gives no information on the jump height of
 the next jumper

$$P(I_{100}=1, I_{111}=1) = P(I_{100}=1)P(I_{111}=1)$$

b) R_i indicator $\rightarrow 1$ i th jumper sets record
 $\rightarrow 0$ " " does not " "

$$\sum_{i=2}^n \mathbb{E}(R_i) = \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

As $n \rightarrow \infty$, the expected # of records set $\rightarrow \infty$